Week 1 web page: delivery guide

Instructor notes for Week1_IntroAndAlgorithms.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week1_IntroAndAlgorithms.html. Open it before class; the three widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week1_IntroAndAlgorithms.R` (same seeds, same numbers). The 100,000-sample salmon loop takes under a second.
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker ch. 1 (printed pp. 1-35; 1.3.2 on p. 9 is the analytical vs. computational distinction) and appendix A (A.1-A.4, pp. 480-482). The schedule also lists ch. 2-3.
- WODs: *Math & Stats Self-Diagnostic* (Monday) and *Pseudocoding a Permutation Test* (Wednesday), keys in `wod/answer_keys/`. Both are worked on the page (`#wod-diagnostic`, `#wod-permutation`).
- Lab 1 Act answers stay off the page. The lab section avoids the linearization of ae^{-bx} (Act 2, 1c), the Ricker derivative and critical point (Act 2, Part 2), the 0-to-2 ramp (Act 2, Part 3), and SSR (Act 3). Its examples use different functions and numbers: a power law, x^2e^{-x}, a 0-to-4 ramp over 5 days.

As taught (Aug. 24-26, 2026)

- The page was built Nov. 1, after the week, from the legacy `INTRO` and `LECTURE1` pages, `NOTES_Week1_Lecture2_AlgorithmsInStatistics`, `NOTES_Lab1_MathRefresher` (both now in `lecture_notes/archive/`), the two WODs, Lab 1, and Bolker ch. 1 and appendix A.
- Monday was the syllabus and the diagnostic WOD only, no content. Per the Monday announcement (added to the page Nov. 4): reading Bolker is the main homework (it won't click until ch. 6), notebooks to lectures and laptops to labs, the course's history and its AI overhaul, absences by email, the Dunning-Kruger curve, and 3Blue1Brown's *Essence of Calculus* for dusting off calculus. All of it is now in `#course` and `#wod-diagnostic`.
- Wednesday: the workflow big picture was planned but not covered live (it opened Week 2 instead). The page keeps a short version in the first H1 as a course map. The bootstrap was a cut-first bonus; no record of whether it was reached.

- New on the page: the three widgets, the power-law example, the course-logistics section, the discussion prompts, and the packaged `permutation_test()` (the legacy version read global variables instead of its own arguments).
- `LECTURE1` and `INTRO` are archived now that this page covers them.

Part 1: What this course is about (Monday, ~15-20 min after the syllabus)

Big picture (~5 min)

- Walk the workflow left to right: problem first, then theories, then models (function + noise), fit, compare, revise. Point out where the semester's topics sit (functions, probability, simulation = MODELS; likelihood, Bayes, MCMC = FIT; AIC, CV = COMPARE).
- "Playing" with data, models, and probability: simulate from your own model and see if you can recover the parameters.

How the course runs (~3 min)

- WOD routine, Acts (alone, then group; wrong first attempts are expected), final project pitches, AI policy (closed in class unless a handout says otherwise), "stop me if it doesn't make sense."

WOD (~10 min, including the neighbor comparison)

- Answers: $x = 5$; $r = \ln(5.5)/8 = 0.213$; $f'(x) = 5(4x^2 - 3x)^4(8x - 3)$, $f'(1) = 25$; $R(t) = 2t^2$, total 50; $P(A \cup B) = 0.7$; two children 1/3; CI (b); p-value (b); `site_data[, 2]`; `for (i in seq_along(values))`; braces around the function body.
- Collect the sheets and look at the self-ratings before Lab 1. Low calculus ratings mean more time on the lab's derivative section.

Discussion (~3-5 min)

- Q1 (CI of 0.42 to 0.61, "95% chance the truth is in there") *Land:* strictly, the 95% belongs to the procedure: in repeated samples, 95% of such intervals cover the true value; this one either does or doesn't. A Bayesian credible interval does carry that reading (given a prior), which comes later in the semester. For most management decisions the distinction doesn't change the action; it matters when someone treats the interval's ends as hard limits.
- Q2 (your own problem and two theories) *Land:* open. Push for a problem stated as a decision or need, and for theories that predict different data; if they predict the same data, no study separates them.

Part 2: Math tools for Lab 1 (lab opening, ~10-15 min)

- A refresher, not a first lesson. Do logs and the product + chain rules; skim integration if time is short (Act 2 Part 3 gives a worked constant-rate example on the handout).
- **Board:** exponent rules, then logs as "what power?"; the growing square, then the rectangle.
- Q (power law: rise per tenfold mass; slope of 1) *Land:* 0.75 per decade (0.477, 1.227, 1.977, 2.727). A slope of 1 means rate is proportional to mass, so the rate per gram is constant; a slope of 0.75 means larger animals use less energy per gram.
- **Widget (growing rectangle):** square mode, $x = 3$. At $dx = 1.5$: exact change 11.25, strips 9, corner 2.25 (20%). At $dx = 0.1$: exact 0.61, strips 0.60, corner 0.01 (1.6%). Ask for a prediction before sliding. Then rectangle mode: shrink h and watch the corner share go to 0.
- Q (corner's share; why drop it) *Land:* the share is $dx/(2x + dx)$, which goes to 0. Divided by dx, the corner contributes dx, which vanishes in the limit, so it doesn't affect the derivative. The exact new area still includes it.
- x^2e^{-x}: $f' = x(2 - x)e^{-x}$, critical points 0 and 2. This is the closest parallel to point stuck groups toward in Act 2 Part 2, without giving the Ricker answer.
- **Widget (accumulation):** start with the constant rate (any rule is exact, area 10). Ramp, left edge, 5 rectangles: 8; right edge: 12; midpoint: 10. Rise-and-fall, left edge, 5 rectangles: 6.38 vs. 7.13 exact; midpoint 7.24. Slide to 100 rectangles.
- Q (left/right/midpoint on the ramp) *Land:* the ramp increases, so the left edge takes the lowest height in each slice (too small) and the right edge the highest (too big). For a straight line, the midpoint's overshoot on one half of each slice exactly cancels the undershoot on the other; a curved function doesn't cancel.
- Circulating: Act 1 and Act 3 need none of this. Act 2 Part 1 needs log rules; Part 2 product + chain; Part 3 "area = total."

Part 3: Algorithms for inference (Wednesday, ~32 min)

Big picture (~3 min)

- COMPARE box, simplest version: data vs. a null model. Analytical (formula, exact, needs assumptions) vs. computational (a program, approximate, needs only the story). Bolker 1.3.2.

WOD (~6 min)

- Statistic: difference in means, $174.8 - 178.2 = -3.4$ g. Two-tailed (no direction predicted). Pseudocode: shuffle labels, recompute, store, compare. Have a couple of students read their steps aloud before showing the page's list; the R version waits for section 4.

Two routes to a z-test (~10 min)

- $\bar{x} = 3.476$, $SE = 0.285$, $z = -3.60$, $p = 0.00016$.

- Simulation: 21 of 100,000 means at or below 3.476, $p \approx 0.00021$. SD of simulated means 0.286 vs. $SE = 0.285$: the loop finds the standard error without the formula.
- **Board pseudocode:** simulate a sample under the null, compute its mean, store; repeat; p = fraction at least as extreme.
- One-tailed here vs. two-tailed for the lizards; worth a check-in.

Permutation test (~10 min)

- $p \approx 0.37$ (seed 746); t-test 0.36. Not significant, a useful contrast with the salmon.
- Put the page's loop next to the WOD's fill-in skeleton.
- **Widget (resampling):** lizards first, 10,000 replicates (about 0.37, the shaded tails on both sides). Then salmon: add 1,000 at a time; most runs show 0 extreme means. Ask what p to report before answering.
- Q (salmon runs of 1,000 with no extreme means) *Land:* report $p < 0.001$ (at most 1 in 1,000), not $p = 0$. With true $p = 0.00016$, 1,000 replicates expect 0.16 hits, so about 85% of runs show none. Seeing about 5 takes roughly 30,000 replicates; the page's 100,000 gave 21.
- `permutation_test()`: packaging the loop. Skip if behind.

Discussion (~3-5 min)

- Q1 (formula 0.00016 vs. simulation 0.00021) *Land:* the simulation's own standard error is about $\sqrt{p(1-p)/N} = 0.00004$, so they agree within noise. Trust the formula when its assumptions hold (here they are given); trust the simulation when the formula's assumptions fail or no formula exists, and add replicates until the noise is small relative to the decision.
- Q2 (when permutation and t-tests disagree) *Land:* small samples with strong skew or an outlier (the t-test leans on normality of the mean), or very unequal spreads. The permutation test assumes only that labels are exchangeable under the null, so it holds up there.
- Q3 (own research) *Land:* open. The random step is whatever the null says is arbitrary: labels, locations, timing, or draws from a simulated process.

Bonus (cut first): bootstrap

- R^2: girth 0.94, height 0.36. 95% bootstrap intervals: girth 0.90 to 0.96, height 0.14 to 0.59. Third flavor of the same loop (resample rows with replacement). Lab 2 Act 3 uses a bootstrap, so don't work its example here.